Two types of alkyl derivatives can be obtained. Those derived by replacement of H – atom of hydrogen cyanide by the alkyl groups are known as alkyl cyanides (R-C ≡ N). and those obtained by the replacement of H – atom of hydrogen isocyanide are known as alkyl isocyanides (R-N ⟹ C)

In IUPAC system, alkyl cyanides are named as "alkanenitriles" whereas aryl cyanides as " arenecarbonitrile".

Table : Nomenclature of cyanides

Compound (common name, Structural formula, IUPAC Name)	IUPAC Name			
	Prefix with position number	Root used	Primary suffix	Secondary Suffix
acetonitrile CH_3-CN Ethane nitrile	–	Eth	ane	nitrile
Propiononitrile CH_3CH_2-CN Propanenitrile	–	Prop	ane	nitrile
Butyronitrile $CH_3CH_2CH_2$-CN butanenitrile	–	But	ane	nitrile
Isobutronitrile CH_3-CH(CH$_3$)-CN 2-methylpropanenitrile	2-methyl–	prop	ane	nitrile
Benzonitrile C_6H_5-CN Benzene Carbonitrile	–	Benzene	Carbo	nitrile
$H_3C-CH(CN)-CH_2-COOH$ 3-Cyanobutanoicacid	3-Cyano	but	ane	oicacid
$C_2H_5-C(CH_3)(Cl)-CH_2(Br)-CN$ 2-Bromo-3-chloro-3- methyl pentanenitrile.	2-Bromo-3-chloro-3- methyl	pent	ane	nitrile

XII U13-Organic Nitrogen Compounds.indd 223 16-12-2022 19:45:27